When Correlation Isn't Enough: Using Causal Inference for Algorithm Bias Detection

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Abstract*—Algorithmic fairness assessments rely predominantly on correlation-based metrics — statistical parity, equalized odds, disparate impact — that detect *whether* a disparity exists but cannot distinguish direct discrimination from effects mediated by proxy variables, nor can they trace the causal pathways through which biased outcomes arise. This limits root-cause analysis, legacy-data vetting, and the targeted sampling decisions needed to validate high-stakes decision systems. We close this gap by benchmarking six causal discovery algorithms — PC, FCI, GES, GRaSP, ICA-LiNGAM, and DirectLiNGAM — on bias detection tasks with known ground truth. On a synthetic loan-approval dataset with planted discriminatory bias, the linear non-Gaussian acyclic functional causal models recover the planted edge more accurately than constraint- and score-based methods. Sensitivity analysis across 480 controlled runs establishes the operating regimens per algorithm over sample size and effect magnitude. Applied to the ProPublica COMPAS recidivism dataset, DirectLiNGAM estimates a positive direct Race-to-Score magnitude coefficient that agrees with the covariate-adjusted direct effect under full criminal-history controls within 0.0088 points on the 1-10 decile scale; bootstrap confidence intervals exclude zero and substantially overlap. We further characterize a failure mode of BIC-driven permutation search (GRaSP) in the fairness-audit setting: algorithmic scores, being constructed from their own predictors, exhibit mechanically low residual variance and are misclassified as exogenous, producing implausible edges directed into immutable attributes. Causal inference complements correlation-based assessment: correlation reveals *that* a disparity exists; causal inference reveals *where* it comes from.

***Index Terms*—causal inference, algorithmic fairness, bias assessment, COMPAS, disparate impact, LiNGAM, structural causal models**

I. Introduction

In this paper, we use methods from causal science—causal discovery and causal inference algorithms—to investigate potential discriminatory biases in sample datasets and address the question: To what extent can causal discovery algorithms identify structural patterns in data that produce biased outcomes? By applying these methods to toy and empirical datasets, we characterize how one variable (or node in a causal graph) influences another and systematically test whether inferred causal relations are genuine rather than artifacts or products of unmeasured confounding. Causal analysis aims to uncover the fundamental mechanisms behind cause–effect relationships. By contrast, probabilistic models typically depend on previously observed data to estimate how likely certain events are to recur (correlation) without asserting any causal relationships or incorporating interventions.

We argue that causal discovery and causal inference algorithms are underused for detecting bias in historical and observational data. These techniques can provide a more detailed mechanistic understanding of how bias is generated than methods based solely on conventional statistical associations. Identifying bias in datasets and the causal pathways through which it arises can help domain experts determine root causes and mitigation strategies.

In today's rapidly evolving technological environment—with growing AI adoption and expanding automation objectives—decision-support systems shape outcomes in bail decisions, credit scoring, hiring, and healthcare on a population-wide scale. When such systems embed historical or structural discriminatory biases, the resulting harms are unevenly distributed: they fall most heavily on groups already experiencing disadvantage, and they do so with a speed and scale that make it difficult for human oversight to intervene effectively and promptly [1], [2].

The fairness-assessment tools most widely deployed today – statistical parity [3], equalized odds [4], and disparate impact ratio (DIR) [5] – are exclusively correlation-based. They answer the question: *is there a disparity?* They cannot answer: *where does it come from?* Without mechanistic understanding, interventions remain guesswork. Removing a sensitive feature does not remove a proxy discriminator [6]. Rebalancing data does not address upstream structural inequities. Causal frameworks have the ability to identify potential interventions using Pearl's *do calculus*. For example, researchers can study the consequences of giving or withholding treatment or implementing an intervention measure [7], [8] while remaining within ethical boundaries.

II. Background

Many public and private sector domains are using assessments to review AI-assisted decision systems for safe, responsible, explainable, and ethical use. An often used example of potential harms is the Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) tool, a recidivism-prediction system used across many U.S. jurisdictions to estimate the likelihood of reoffending for sentencing decisions [1]. In ProPublica's Broward County analysis, among defendants who did not reoffend within two years, 44.9% of African-American defendants (shown as false positives, FP) were labeled medium/high risk versus 23.5% of Caucasian defendants, implying a selection-rate disparity of about 1.91 between the two groups [9]. This statistic by itself cannot reveal whether race affects the scores *directly* (meaning the scoring rules explicitly embed impermissible racial effects and thus require revision) or *indirectly* through prior convictions, a proxy discriminator that is elevated due to possible racially biased policing (pointing instead to the need for earlier-stage policy reforms). Assessment methods based solely on correlations cannot disentangle these alternative causal routes; methods grounded in causal inference, in contrast, can distinguish between them. Our contributions can be summarized as follows:

1) **A new approach using causal analysis with a proposed pipeline to detect bias (Fig. 6).** We use six algorithms of different types (shown in Table I) with synthetic data that has a known ground truth, and we systematically characterize which methods reliably recover planted discrimination paths and quantify their magnitude in Section V. We then turn to a reportedly biased dataset from the real-world to see if our approach will detect bias, identify the variables that influence the outcome, determine the influential magnitude between variables, and finally outline the causal pathway. We also show ways to validate the bias using convergent estimates, an adjusted direct-effect estimate, and bootstrapping.
2) **Real-world COMPAS application.** Two of the six algorithms independently identify a direct causal pathway from Race (represented by the oriented edge $\rightarrow$) to Score. Moreover, DirectLiNGAM and ICA-LiNGAM yield convergent estimates of the corresponding causal coefficient, which is somewhat expected due to their underlying methodology exploiting non-Gaussianity as functional causal models. This result serves to cross-validate the finding. The adjusted direct effect (DE) further corroborates this result in Section VI using this empirical dataset.
3) **Permutation-based algorithm failure in the audit setting.** We characterize how a known hazard of residual-variance ordering heuristics [10], [11] manifests structurally in algorithmic-fairness audits. Because an algorithmic score is constructed from its own recorded predictors, it is necessarily highly explained by the remaining variables (on COMPAS, the outcome has the highest $R^2 = 0.45$), and a BIC-driven permutation search such as GRaSP (Greedy Relaxations of the Sparsest Permutation) [12] misclassifies the score as exogenous, producing graphs with arrows directed into immutable demographic attributes. We observe this failure in two regimes — at low sample sizes ($n \leq 1{,}000$) on synthetic data, where it attenuates as n grows, and on real-world scores regardless of sample size (COMPAS, $n{\approx}5{,}000$), where it is driven by the score's constructed nature rather than sampling noise. We provide both the analytical mechanism and empirical confirmation across the sensitivity grid in Section V-D and on the COMPAS dataset in Section VI.

III. Related Work

Correlation-based fairness and its limits. As previously noted, dominant tools in correlation-based fairness bias detection include statistical parity [3], equalized odds and equal opportunity [4], the disparate impact ratio (DIR) [5], and adversarial debiasing [13]. These *methods detect that* a disparity exists but do not *explain why*. The DIR (also known as "the four-fifths rule") is the correlation-based baseline against which causal estimates are compared in Section VI.

Causal fairness, discovery, and inference. Kusner et al. [14] formalizes counterfactual fairness within structural causal models (SCM); Goethals et al. [15] introduce Predictive Counterfactual Fairness (PreCoF) for counterfactual explanations; Su et al. [6] survey path-specific methods. Glymour et al. [16] survey graphical-model-based causal discovery. Reisach et al. [10] show that variance-ordering heuristics in causal discovery are sensitive to the alignment of marginal variance with causal order (varsortability); we extend this line of analysis to algorithmic-score auditing, where the hazard is structural rather than incidental. Shimizu et al. introduce ICA-LiNGAM [17] and DirectLiNGAM [18]. Cheng et al. [19] advocate causal learning for socially responsible AI. We complement these works by focusing on the *discovery* step – learning which structural paths encode bias **before** mitigation – and by employing Pearl's backdoor criterion to estimate the ATE [8] on the synthetic data as an independent causal-inference estimator that cross-validates the LiNGAM findings.

IV. Methodology

A. Causal discovery algorithms

Causal discovery algorithms, including those employed in this study, are instrumental in identifying which variables and causal pathways exert influence on other variables, thereby exposing the underlying cause–effect relationships. Table I summarizes the six algorithms used in our causal analysis. PC [20] and FCI [21] use Fisher-Z conditional independence tests (α=0.05) and orient edges via v-structures. FCI additionally generates bidirected ($\leftrightarrow$) edges to represent latent common causes (confounders). We chose the Fisher-Z test because the synthetic data-generating process is linear with continuous variables, and Fisher-Z is computationally more efficient than the Kernel Conditional Independence (KCI) test alternative, which is computationally impractical at the large sample sizes used in the sensitivity analysis.

GES [22] conducts a two-stage greedy search under the Bayesian Information Criterion (BIC), exploring graph structures that maximize the score. GRaSP [12] offers a theoretical improvement over GES in the finite-sample regime. ICA-LiNGAM [17] and DirectLiNGAM [18] leverage non-Gaussianity; the latter relies on sequential regression, guarantees convergence, and is more robust to binary as well as heavy-tailed variables that are typical of high-stakes administrative data. The LiNGAM variants also conveniently quantify the strength of directed edges. For instance, a Race$\rightarrow$Loan coefficient of $\beta = -0.15$ represents the standard deviation

TABLE I
CAUSAL DISCOVERY ALGORITHMS AND PROPERTIES

Algorithm	Family	ID Latent confounder	Unique dirs.	Identifies $\hat{\beta}$
PC	Constraint	No	Partial	No
FCI	Constraint+	Yes	Partial	No
GES	Score (BIC)	No	Partial	No
GRaSP	Permutation	No	Partial	No
ICA-LiNGAM	Func. causal	No	Full	Yes
DirectLiNGAM	Func. causal	No	Full	Yes

decrease in the probability of loan approval for minority applicants. We therefore use the LiNGAM variants to capture this magnitude effect. All experiments are implemented with the `causal-learn` package on both synthetic and real-world datasets [23].

B. Sensitivity analysis of causal discovery algorithms

To supplement the synthetic data beyond a tool to detect bias, we asked *what an ideal or adequate sample size is for each of the algorithms, and at what magnitude (β) each algorithm reliably detects bias?* Since we control both the biased ($\beta = -0.15$) and unbiased ($\beta = 0.0$) parameters, this gives us additional insight into selecting the right algorithm for the task at hand: detecting bias in a dataset. It establishes algorithmic reliability beyond a single experimental condition using this method. It uses the bias coefficient and sample size across 480 controlled runs, reporting the detection rate and the mean SHD ($\pm$) standard deviation per cell.

C. Evaluation metrics

Structural Hamming Distance (SHD). For the synthetic dataset, the Structural Hamming Distance (SHD) uses a binary detection (true/false) of the planted bias edge and compares it to the known ground truth. The SHD measures how many edges are added, deleted, or reversed to align with the ground truth graph. For this metric, the lower the SHD, the more closely aligned the graph is with the ground truth, with zero indicating that they are identical.

The six algorithms return different graph objects — DAGs (ICA and DirectLiNGAM), CPDAGs (GES, GRaSP), and PAGs (FCI) — which we score against the ground-truth DAG under a single convention: a predicted directed edge matching the true edge counts as correct; a reversed edge counts as one error; a spurious or missing edge counts as one error; and an undirected or partially oriented prediction counts as one error, even when the true edge exists in either direction. This convention deliberately penalizes unoriented output because the audit task requires orientation — an auditor cannot act on Race—Score without knowing its direction. We note the trade-off: algorithms that correctly represent orientation uncertainty (PC, FCI) are penalized relative to algorithms that commit to a direction, so SHD here measures the fitness for the audit task rather than correctness relative to each algorithm's own identifiability class.

Causal effect estimation (ATE and direct effects). To quantify the causal effect of selected variables on designated outcomes (e.g., Race$\rightarrow$Loan), we employ causal inference methods based on Pearl's backdoor criterion [7]. Specifically, we estimate the ATE of Race on the outcome using linear regression, conditioning on all non-descendant variables in the inferred directed acyclic graph (DAG). The causal estimation procedure is implemented using the DoWhy library [24]. The formula for the ATE is applied to the biased dataset (Minority = 1), Dataset A.

$$\text{ATE} = E[Y_i(1) - Y_i(0)] \tag{1}$$

In Study 2, where Priors mediates Race$\rightarrow$Score, adjustments for criminal history target the controlled direct effect rather than the total effect (Section VI-C).

Disparate Impact Ratio (DIR). In order to provide an equal comparison between the correlation measure and the results of the causal discovery algorithms, we use the correlation-based DIR steps [5] and the following formula:

$$\text{DIR} = \frac{P(\hat{Y} = 1 \mid \text{Protected})}{P(\hat{Y} = 1 \mid \text{Favored})} \tag{2}$$

where $\hat{Y} = 1$ denotes the decision outcome of interest (e.g., loan approved or classified as high-risk), and the conditioning distinguishes the protected from the favored group. It should be noted that in one experiment (Section VI-A), this formula is algebraically inverted to provide researchers with appropriately formatted results for comparative analysis.

Bootstrap validation. We select 1,000 nonparametric bootstrap resamples of the COMPAS analysis frame ($n = 5,278$ rows sampled with replacement; deterministic per-iteration seeds derived from a fixed base seed). On each resample, the full estimation is repeated from scratch: the adjusted direct effect is re-estimated by OLS, and each LiNGAM variant is refit without background knowledge, with the Race$\rightarrow$Score coefficient recorded as zero whenever a resample's recovered graph omits or reverses the edge — so structural instability widens rather than artificially narrows the intervals. The 95% confidence intervals are the 2.5th and 97.5th percentiles of the resulting distributions and therefore reflect both sampling variability and uncertainty in the discovered structure. The unconstrained bootstrap point estimate is consistent with the constrained full-sample estimate (Section VI-B), and its interval contains both.

Additional metrics. The LiNGAM coefficient $|\beta|$ shows the strength or magnitude of a parent variable on a child variable (linearity). In addition to bootstrapping, we also used cross-algorithm convergence on the COMPAS dataset. We perform the E-value computation using VanderWeele and Ding's approximation for continuous outcomes [25].

V. Study 1: Synthetic Loan-Approval Dataset

A. Data-generating process and terms

We generate two paired datasets (n=5,000) with seven variables: Race, Gender, Education, ZIP Code, Income, Credit

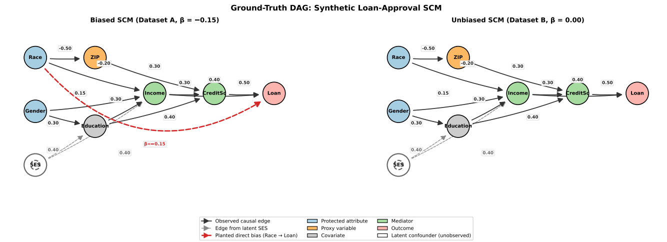

Fig. 1. Ground-truth DAG for both the biased (A) and unbiased (B) datasets. Dashed red edge = planted direct discrimination (β=-0.15). Node colors: blue = protected attribute, orange = proxy, green = mediator, pink = outcome. The dotted red edge is absent in Dataset B (unbiased) as expected.

Score, and Loan approval. We set the bias coefficient to β=-0.15. This β value is a magnitude consistent with documented bias across multiple algorithmic decision domains, and it aligns with the discrimination magnitude that researchers have indicated in real-world lending datasets: Bertrand and Mullainathan (hiring callbacks – 10 pp gap corresponds to roughly $\beta \approx -0.10$ on a standardized outcome [26]); Obermeyer et al. (healthcare risk scores – $\sim 0.21\sigma$ gap corresponds to $\beta \approx -0.21$ [27]); and Bartlett et al. (mortgage approvals – 3–8 pp gap corresponds to $\beta \approx -0.05$ to -0.10 [28]). Our chosen $\beta = -0.15$ sits between these and is calibrated to be detectable across all six algorithms, while the full operating range is described in Section V-D.

All variables are standardized to a range (-2, +2). Dataset A (biased) includes a direct Race$\to$Loan edge with β=-0.15. Dataset B (unbiased) sets β=0. A hidden socioeconomic-status (SES) variable provides a latent confounder detectable by FCI. We use standard fairness terminology: Race and Gender are protected attributes; ZIP Code is a proxy discriminator — a variable that appears neutral but results in biased outcomes; Income and CreditSc are mediators. In the following DAG, which indicates a chain structure, a mediator is illustrated by the M variable in the following chain and acts as a bridge between variables:

$$X \longrightarrow M \longrightarrow Y$$

Figure 1 shows the ground-truth DAG for Dataset A. The dashed red edge encodes direct discrimination; the proxy path Race$\to$ZIP$\to$CreditSc$\to$Loan is present in both datasets.

B. Results from causal discovery algorithms

Table II summarizes all six algorithms. Both LiNGAM variants and GRaSP detect the Race$\to$Loan edge in Dataset A (biased) and correctly omit it in Dataset B (unbiased). Both LiNGAM variants recover $\hat{\beta}$=-0.1074 within 0.05 of the planted -0.15, providing a magnitude estimate that GRaSP does not. GRaSP's detection at this sample size comes at the cost of a higher structural error on the unbiased dataset (SHD = 11), discussed further in Section V-E and in the Discussion VII. FCI correctly surfaces an undirected Education to Income edge, signaling the hidden socioeconomic status (SES) confounder, shown as the white node SES in Figure 1. No background knowledge was provided to any algorithm in Study 1.

TABLE II
CAUSAL DISCOVERY ALGORITHM PERFORMANCE ON THE SYNTHETIC DATASET (N = 5,000)

Algorithm	**SHD** Bias	**SHD** Unbias	**Race→Loan** Bias	**Race→Loan** Unbias	**Lat. conf.**	$\hat{\beta}$ **(bias)**	$\hat{\beta}$ **(unbias)**
PC	8	7	No	No	No	–	–
FCI	9	8	No	No	Yes	–	–
GES	13	3	No	No	No	–	–
GRaSP	3	11	Yes	No	No	–	–
ICA-LiNGAM	1	1	Yes	No	No	-0.1074	0.0
DirectLiNGAM	1	1	Yes	No	No	-0.1074	0.0

SHD lower = better. Race$\to$Loan edge detected for GRaSP, and the LiNGAM variants which both have low SHD results indicating near perfect alignment with the ground truth.

C. Backdoor adjustment ATE on the synthetic dataset

Using linear regression on the biased synthetic dataset, applying the backdoor adjustment {Income, CreditSc, ZIP, Gender, Education}:

$$\begin{aligned} \text{ATE}_{\text{no controls}} &= -0.2967 \quad \text{(raw correlation)} \\ \text{ATE}_{\text{ZIP+Income+CreditSc}} &= -0.1088 \quad \text{(partial controls)} \\ \text{ATE}_{\text{full controls}} &= -0.1086 \quad \text{(full controls)} \end{aligned}$$

The ATE drops substantially as controls are added, but it does *not* reach zero. This is interpreted as supporting the detection of bias with full controls applied; the ATE indicates that minority applicants are estimated to have a -0.1086 standard deviation reduction in loan score due to direct racial discrimination, with a standard error (SE) of $+0.0182$ after controlling for Gender, Education, ZIP, Income, and Credit Score. For reference, the raw standardized mean difference in loan score between groups is -0.2967σ (equivalently, the no-controls estimate above). We note that the DIR is undefined for a continuous standardized outcome — selection-rate metrics require a binary decision threshold — which illustrates the limits of correlation-based metrics for continuous scores.

D. Sensitivity experiment method, results, and graph interpretation

A key limitation of the initial synthetic experiments is that they only perform a single experiment at fixed n and fixed β. To establish algorithm operating regimens and to provide practical guidance on when each algorithm method is more reliable or trustworthy at different sampling levels, we use $\beta \in \{0.00, 0.05, 0.10, 0.15, 0.20, 0.25\}$ across sample sizes $n \in \{1000, 5000, 10000, 50000\}$ with 20 random seeds per cell (6 betas x 4 sizes x 20 seeds = 480 runs total). For each cell, we report: (1) *detection rate* — the fraction of bias detection executions where Race$\to$Loan is present in the recovered graph (for biased conditions) or absent (for unbiased); and (2) *mean SHD* $\pm$ *standard deviation*. We measure $|\beta|$ for detection; signs are dataset-specific: negative reduces minority loan-approval likelihood in Study 1; positive inflates the COMPAS score for African-American defendants in Study 2 (interpreted as an increased risk for recidivism).

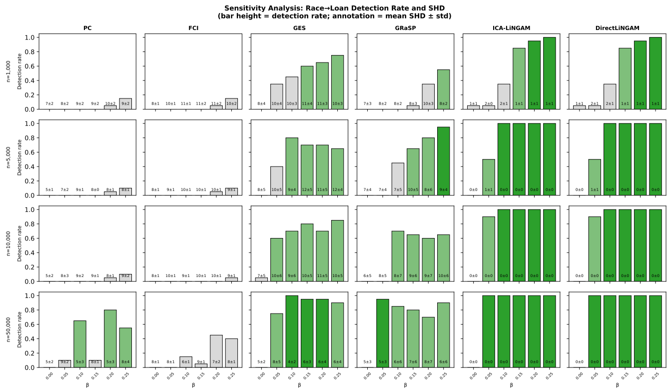

Fig. 2. Sensitivity analysis: Race→Loan detection rate (color intensity) and sample size n (rows), across all algorithms (columns). The bar height encodes detection rate; the annotation within each bar reports the mean SHD ± standard deviation over 20 seeds, capturing the structural accuracy of the recovered graph independently of whether the bias edge was detected.

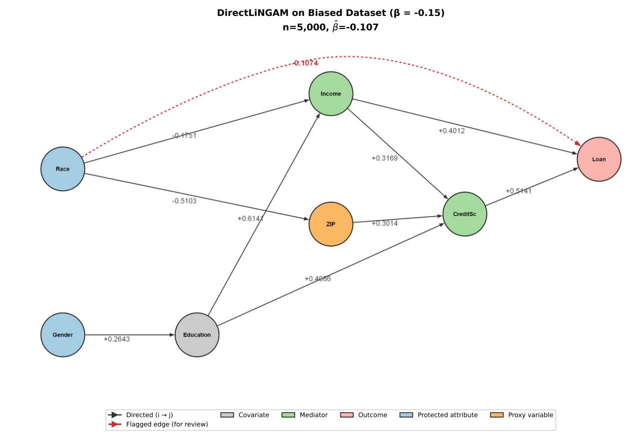

Fig. 3. Causal discovery results using DirectLiNGAM on the synthetic biased dataset with the planted $\beta = -0.15$. Both LiNGAM variants recover the Race→Loan edge (in red), with a returned value of -0.1074, within 0.05 of the planted β.

E. Key findings from the sensitivity analysis

Fig. 2 reports the detection rate and SHD across (n, β) cells. The fourth bar of each cell corresponds to the main experiment's planted $|\beta| = 0.15$ (or magnitude 0.15). This sensitivity analysis reveals a striking stratification among algorithm families.

Finding 1: Constraint-based methods fail. PC and FCI failed to reliably detect direct discrimination at any combination of sample size and effect magnitude tested, with detection rates below 0.50 even at $n = 50,000$ and $\beta = 0.25$ for FCI, and structural error rates (SHD $\approx$ 5–11) indicating a systematic misorientation of the Race→Loan edge. We note that PC and FCI return equivalence classes; their non-detection partly reflects that the Race→Loan orientation is not identifiable from the conditional independencies alone, so "failure" here means failure for the audit task, which requires orientation on the edges.

Finding 2: Score-based and permutation-based methods are marginal but improve with sample size. GES achieved moderate detection at lower sample sizes as β increased, reaching approximately 70% at $\beta \geq 0.10$ and $n \geq 5{,}000$. GRaSP detection rate is near-zero at $n = 1{,}000$ across all magnitudes tested, but it improves substantially at $n \geq 5{,}000$: at our main experimental setting ($\beta = -0.15$, $n = 5{,}000$), GRaSP recovers the planted Race→Loan edge with low structural error SHD $= 3$ (Table II). However, on the unbiased dataset with the same sample size, GRaSP produces a substantial false-positive structure (SHD $= 11$), indicating high variance across seeds and limited reliability as a single-decision assessment tool. GES and GRaSP also exhibited higher residual structural errors at lower sample sizes (SHD $\approx$ 8–12).

Finding 3: Functional causal models reliably recover the planted edge. ICA-LiNGAM and DirectLiNGAM achieved $\geq$ 85% detection at $n = 1,000$ and $\beta = 0.15$, near-perfect detection ($\geq$ 95%) at $n = 5,000$ for $\beta \geq 0.10$, and SHD $\approx$ 0–2 throughout. Constraint- and score-based methods systematically fail to detect effects of the magnitude reported in empirical discrimination research [1] across all sampling sizes tested; therefore, algorithm choice is a substantive methodological decision, not a neutral one. Fig. 3 shows the DirectLiNGAM result on the synthetic data with the oriented Race→Loan bias edge.

VI. Study 2: COMPAS Recidivism Dataset

In the synthetic experiments, β denoted the structural equation coefficient planted directly into the data-generating process, representing the precise causal penalty imposed on minority applicants through the direct Race → Loan path. This known ground truth enables our rigorous evaluation of each algorithm's detection accuracy and coefficient recovery.

In the COMPAS analysis, $\hat{\beta}$ denotes LiNGAM's estimated direct causal effect of race on the COMPAS score under the assumptions of the linear non-Gaussian acyclic model. Unlike the synthetic case, no ground truth is available for comparison, and the estimate is subject to bias from unmeasured confounders.

A. COMPAS dataset and preprocessing

The COMPAS dataset [1] records defendants screened in Broward County, Florida, during the years 2013–2014. We apply the standard ProPublica preprocessing filters (|days_b_screening_arrest|≤30, is_recid $\neq -1$, charge degree $\in \{F, M\}$, score_text $\neq$ N/A) on the publicly available `compas-scores-two-years.csv` to n=6,172. Then we restrict the dataset to include only the African-American and Caucasian defendants reported by ProPublica, yielding n=5,278 observations (3,175 African-Americans and 2,103 Caucasians). Nine variables are encoded: Race (African-American=1, Caucasian=0), Sex (Male=1), Age, Juvenile Felony Count, Juvenile Misdemeanor Count, Prior Convictions, Charge Degree (Felony=1), COMPAS Decile Score (1–10), and Recidivism (`two_year_recid`, 0/1). On this sample, the two-year recidivism rate differed by race (Table III), indicative of a baseline disparity. African-American (AA) defendants received a mean COMPAS decile score of 5.28 versus 3.64 for Caucasians (Cau), a raw mean-score gap of

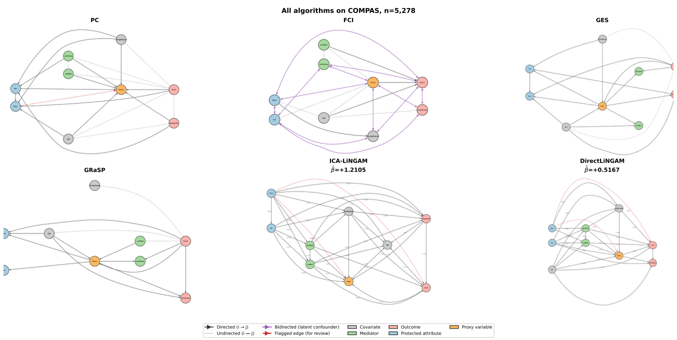

Fig. 4. Causal graph results from the six algorithms applied to the COMPAS dataset.

1.64 (descriptive). The DIR, created and used by employment-discrimination case law (four-fifths rule), is defined for binary selection rates, not means of ordinal scores [5]; we therefore report it as the ratio of high-risk classification rates:

$$\begin{aligned} &P(\text{Score}{\geq}5 \mid \text{AA})/P(\text{Score}{\geq}5 \mid \text{Cau}) \\ &\quad = 0.576/0.331 = 1.741 \end{aligned} \tag{3}$$

This equation is inverted to compare against the four-fifths threshold; the protected-over-favored ratio is $0.331/0.576 = 0.575 < 0.80$, indicating a disparate impact under the conventional definition. The recidivism rate ratio $0.5231/0.3909 = 1.338$ is similarly above parity. These correlation-based metrics demonstrate the existence of disparities; however, they do not localize the underlying sources of these differences. This limitation motivates our subsequent causal discovery analysis.

TABLE III
TWO-YEAR RECIDIVISM RATES ON THE PROPUBLICA-FILTERED DATASET (n=6,172).

Race	n	Recidivated	Rate
African-American	3,175	1,661	**52.31%**
Caucasian	2,103	822	**39.09%**
Other	894	326	36.47%

Rates are computed from `two_year_recid` after applying the standard ProPublica filters; Section VI restricts the experiment to the African-American and Caucasian defendants (n=5,278).

B. Causal discovery results on COMPAS data

All algorithms received the same nine-variable frame on its natural scales (decile score 1-10, binary Race/Sex/ChargeDegree, raw counts); no standardization is applied, so all reported effects are in decile-score points. DirectLiNGAM was run with prior background knowledge encoding two domain constraints: the immutable demographic attributes (Race, Sex, Age) were declared exogenous (no incoming edges), and Recidivism, which post-dates the score, was declared a sink. Encoding such constraints is standard practice for DirectLiNGAM [18] and reflects knowledge that no dataset can overturn — nothing in the data causes a defendant's race.

A robustness run without any background knowledge recovers the same Race→Score edge with $\hat{\beta} = +0.4834$, within 0.034 of the constrained estimate, indicating that neither the detection nor the magnitude is an artifact of the constraints.

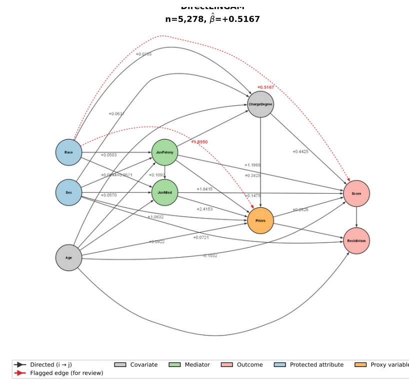

Fig. 5. DirectLiNGAM applied to COMPAS (n=5,278). The dashed red edge is the detected bias path, Race→COMPAS Score $\hat{\beta}$=+0.5167. The Race→Priors path $\hat{\beta}$=+0.1895 shows the proxy discriminator, Priors, adding further to the COMPAS score for African-Americans.

The unconstrained graph does, however, contain temporally impossible orientations (e.g., Recidivism→Score and edges directed into Race), which the constraints exist to prevent; notably, even unconstrained, DirectLiNGAM's causal order places the Score last, whereas GRaSP places it at the root. Full unconstrained output is provided in the replication repository.

The remaining five algorithms, including ICA-LiNGAM, ran unconstrained, as causal-learn supports prior knowledge only for DirectLiNGAM among the six. Cross-algorithm orientation comparisons should be read accordingly: DirectLiNGAM's absence of edges into immutable attributes is partly by construction, whereas ICA-LiNGAM's recovery of Race→Score and Race→Priors was obtained without constraints and provides independent support for the edge's existence. The detection of Race→Score by DirectLiNGAM remains informative — the constraints forbid arrows into Race but do not require any arrow out of it.

The resulting DAGs are shown in Fig. 4. **PC** produces several edge orientation reversals regarding the immutable characteristics of Race and Sex; however, they are shown to be caused by other nodes, consistent with its known limitations under strong mediation. However, PC does correctly orient Race→Priors indicated by the red dotted edge. **FCI** surfaces a high number of bidirectional (↔) edges indicating latent common causes — but does not recover a correctly oriented Race→ Score edge.

GES and **GRaSP** do not recover a correctly oriented Race→COMPAS Score edge; instead, both orient it as Score→Race, along with a number of other implausible edges, indicating failure. Both **ICA-LiNGAM and DirectLiNGAM** methods recover the causal relationships between Race→COMPAS Score and Race→Priors. However, a closer inspection reveals incorrectly oriented edges into Age in the ICA-LiNGAM results, which is a known cost of running unconstrained. Figure 5 shows the DirectLiNGAM graph with annotated $\hat{\beta}$ coefficients.

C. Adjusted Direct Effect of Race on COMPAS Score

Our causal question concerns *direct* discrimination: does Race influence the COMPAS score through a pathway not mediated by criminal history? Because Prior Convictions lie on the mediated pathway Race→Priors→Score (Fig. 5), the choice of adjustment set determines which causal quantity is estimated. Conditioning on no covariates targets the **total effect** of Race on Score—the combined influence transmitted through all pathways, both direct and mediated. Conditioning on the criminal-history variables (Age, Charge Degree, Juv_Fel, Juv_Misd, Priors) deliberately blocks the mediated pathways and targets the *controlled direct effect* (CDE) of Race on Score [7], [29]. Under our linear, no interaction structural model, the CDE is constant across mediator levels and coincides with the natural direct effect [29]; we therefore refer to it simply as the *adjusted direct effect*. We note that this estimand—not the total effect—is the appropriate target for the direct-discrimination hypothesis, and it is the same quantity as the LiNGAM edge coefficient $\hat{\beta}$ (Race→Score) estimates, which allows for the comparison noted in Section VI-D.

Setting T (Race) = 1 for African-American defendants, the direct effect under linearity is estimated as the Ordinary Least Squares (OLS) coefficient on T after adjusting for Z:

$$\mathrm{DE} = \mathbb{E}_Z\Big[\, \mathbb{E}\left[Y \mid T{=}1, Z\right] - \mathbb{E}\left[Y \mid T{=}0, Z\right] \,\Big] \tag{4}$$

Identification requires that Z blocks all backdoor paths into Race→Score and that there is no unmeasured confounding of the mediator-outcome relationships (sequential ignorability [29]); we assess the robustness to violations of this assumption via the E-value analysis in Section VII. Estimation uses linear regression on the natural variable scales, with Race binary (African-American = 1); effects are in decile-score points per group contrast.

- Effect_no controls = $+1.6416$ (total association, all pathways)
- Effect_age + charge = $+1.1386$ (partial adjustment)
- DE_full history = $+0.5255$ (adjusted direct effect)

The estimate attenuates substantially as criminal-history mediators are blocked, confirming that much of the raw score gap is transmitted through and is partly attributable to criminal-history pathways. However, an adjusted direct effect of $+0.5255$ decile points ($\approx$0.18 standard deviations of the score) remains: African-American defendants received systematically higher COMPAS scores than Caucasian defendants who are observationally identical on all available criminal-history covariates. Under the stated identification assumptions, this residual race-associated direct pathway is consistent with the presence of direct bias in the scoring process; it does not, by itself, establish that the proprietary scoring function explicitly encodes race, a distinction we return to in Section VII.

D. COMPAS causal discovery algorithm summary

Table IV summarizes the results. Two of the six algorithms, the LiNGAM variants, independently detect Race→COMPAS Score, with convergent rather than identical magnitude estimates (see Discussion, Section VII).

TABLE IV
CAUSAL DISCOVERY ALGORITHM RESULT COMPARISON ON THE COMPAS DATASET.

Alg.	Type	Race→Score	Race→Priors	Lat. conf.	$\hat{\beta}$
PC	Constraint	No	Yes	False	–
FCI	Constraint+	No	No	True	–
GES	Score (BIC)	No	No	False	–
GRaSP	Permutation	No	No	False	–
ICA-LiNGAM	Func. causal	Yes	Yes	False	+1.2105
DirectLiNGAM	Func. causal	Yes	Yes	False	**+0.5167**

VII. Discussion

Fig. 6 provides an overview of the complete five-stage analytical pipeline introduced and empirically validated in this study. Step 1 identifies the presence of a disparity; Step 2 selects an appropriate family of algorithms capable of localizing its origin; Step 3 delineates the causal discrimination pathway; Step 4 quantifies its magnitude under controlled conditions; and Step 5 assesses its robustness. The remainder of this section examines the empirical findings associated with each step and discusses their implications for practitioners.

On our synthetically-generated dataset with planted bias, we controlled β at a realistic magnitude while preserving detectability across all six algorithms. The LiNGAM variants further provided signed magnitudes for each oriented edge, supporting interpretation as a positive or negative penalty rather than mere presence or absence.

Convergent evidence. The Race→COMPAS Score finding rests on four convergent lines of evidence (Fig. 6, Steps 4–5). We characterize this as convergent rather than independent evidence: all estimators share the same sample and a linearity assumption, though they differ in their identification strategies (non-Gaussianity vs. covariate adjustment). ICA-LiNGAM and DirectLiNGAM — functional causal models using different estimation strategies (ICA decomposition vs. sequential regression) — both recover a direct Race→Score edge. The covariate-adjusted direct effect (SE = 0.0638), derived from OLS on the same data, targets the same estimand as the DirectLiNGAM edge coefficient and agrees with it within Δ = 0.0088 decile-score points. None of the three 95% confidence intervals includes zero, and the DirectLiNGAM and DE intervals exhibit substantial overlap. The E-value is computed following VanderWeele and Ding's approximation for continuous outcomes [25]: the adjusted direct effect is expressed as a standardized mean difference (d = 0.184), converted to an approximate risk ratio via $RR \approx \exp(0.91 \times d) = 1.18$, yielding $E = RR + \sqrt{RR(RR-1)} \approx 1.65$ for the point estimate and $\approx$ 1.52 for the lower confidence bound (evalue_check.py in the replication repository). An unmeasured confounder would thus need risk-ratio associations of at least this magnitude with both Race and COMPAS Score, beyond the measured

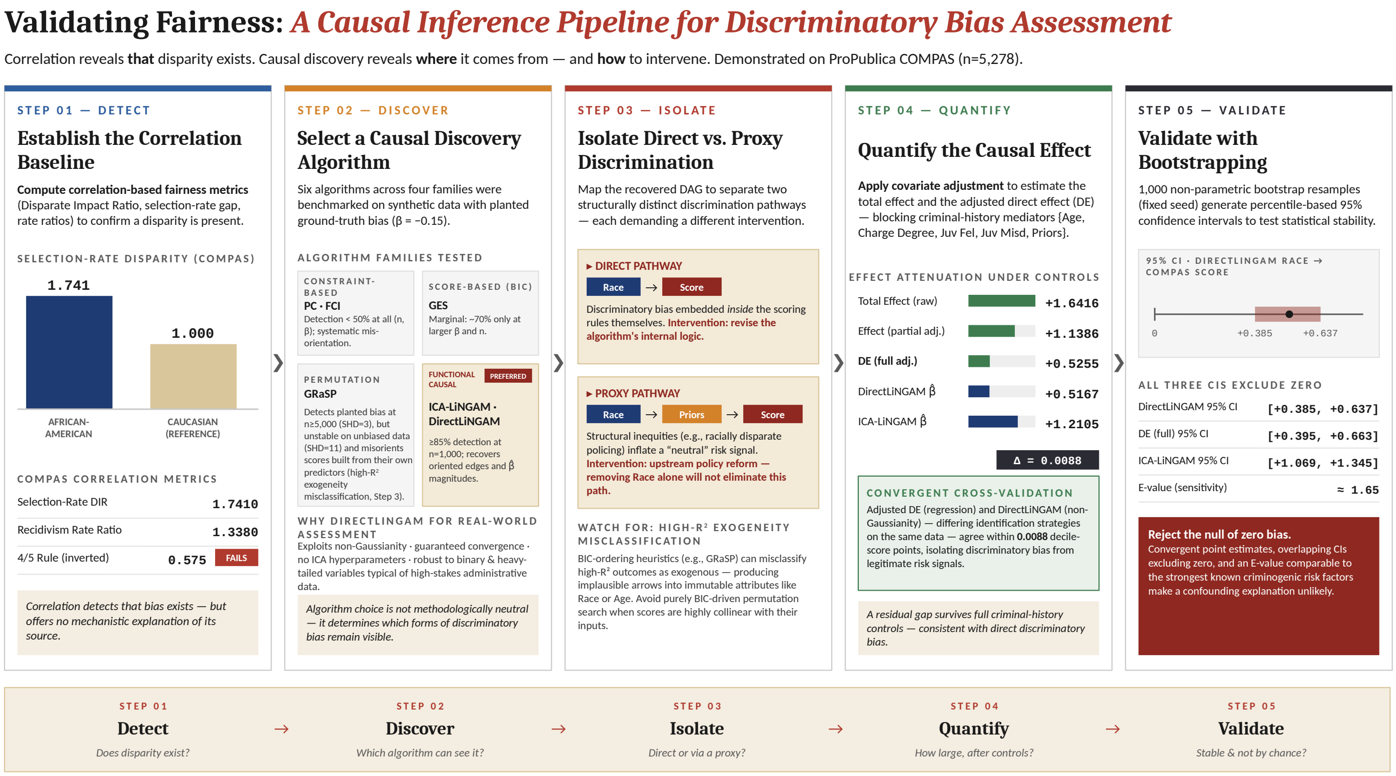

Fig. 6. Five-step causal inference pipeline for discriminatory bias assessment, with COMPAS as the worked example. Correlation-based metrics in Step 1 **Detect:** confirms a disparity. Step 2 **Discover:** uses sensitivity analysis to determine which algorithms to use on the targeted dataset(s). Step 3 **Isolate:** isolates the distinct discrimination pathways (direct vs. proxy discrimination) including a warning about high-R^2 exogeneity misclassification. This step will help guide the appropriate interventions. Step 4 **Quantify:** quantifies the causal effect using the total effect and the adjusted direct effect (DE). Step 5 **Validate:** uses bootstrapping and resampling to ensure the causal effects are not by chance and validate stability.

criminal-history covariates, to fully explain away the direct effect.

DirectLiNGAM serves as a robustness check. On the synthetic biased dataset, ICA-LiNGAM and DirectLiNGAM return identical coefficient estimates ($\hat{\beta} = -0.1074$), within 0.0426 of the planted -0.15, demonstrating the theoretical correctness of both methods when their linearity and non-Gaussianity assumptions are fully satisfied. On the COMPAS data, where discrete count variables (priors, juvenile records) and binary outcomes partially violate these assumptions — the two variants yield convergent but non-identical estimates ($+1.2105$ vs. $+0.5167$). The sign agreement and order-of-magnitude consistency between the two estimates, and with the adjusted DE, constitute meaningful cross-validation. DirectLiNGAM's advantages — guaranteed algorithmic convergence, no ICA hyperparameters, and greater robustness to binary and heavy-tailed variables — make it the preferred variant for real-world assessment tasks (Fig. 6, Step 2).

Direct vs. proxy discrimination: intervention implications. As Fig. 6 (Step 3) shows, the direct Race→Score and proxy Race→Priors→Score pathways demand structurally different interventions. The direct pathway is consistent with race-related adjustments in the scoring process that are not explained by observed risk factors; if substantiated, the indicated intervention is a revision of the scoring rules. The proxy pathway reflects an upstream problem: racially disparate policing may inflate prior-conviction counts, which the algorithm then legitimately uses as a risk signal. Critically, removing Race from the COMPAS model does not eliminate the proxy pathway; addressing it requires upstream intervention in policing practices or in how priors are weighted.

High-R^2 exogeneity misclassification in variance-ordering search. GRaSP's ordering heuristic treats the variable with the *lowest residual variance* after conditioning on all others as "most exogenous". Pathologies of residual-variance-based orderings are known in the causal discovery literature: Reisach et al. [10] show that when a variable's marginal variance is strongly aligned with its causal ordering ("varsortability"), variance-driven orderings can be gamed or inverted. The equal-variance identifiability results [11] make it explicit that those orderings are only valid under variance assumptions that real data do not need to satisfy. Our contribution is to identify when and how this pathology manifests in specific settings of algorithm-fairness audits and to show it is not a one-off or edge case, but a structural hazard. An algorithm risk score is, by construction, a deterministic-plus-noise function of its own recorded inputs; its residual variance given those inputs is therefore mechanically low, and a variance-ordering search will tend to place it first in the causal order. The result is a graph in which the score is rendered exogenous and arrows are directed from the score into immutable demographic attributes. We observe this mechanism in two distinct scenarios or use cases. *First*, at low sample sizes ($n = 1{,}000$) on the synthetic data, finite-sample noise in residual-variance estimates causes GRaSP to misclassify outcome variables as exogenous; this failure attenuates by $n \geq 5{,}000$ as variance estimates stabilize, and GRaSP recovers the correct Race→Loan orientation (Table II).

Second, independent of sample size, the failure recurs on COMPAS ($n = 5{,}278$), where the Decile COMPAS Score has the highest R^2 against the remaining variables ($R^2 = 0.45$; results/compas_r2.csv), precisely because the score is constructed from those variables. GRaSP misreads this multicollinearity as exogeneity, yielding implausible edges such as Score→Race and Priors→Race; the BIC slightly prefers this impossible structure (ΔBIC $= 4.6$), leaving the search in a local optimum. Arrows directed into immutable attributes are thus a cheap and effective diagnostic: any recovered graph containing them should disqualify a purely BIC-driven ordering or permutation search for that dataset. The practical implication for fairness audits is categorical rather than statistical — algorithmic scores are *always* highly explained by their own input features, so this setting sits permanently inside the failure mode that the variance-ordering literature identifies.

Limitations. LiNGAM assumes linear relationships and non-Gaussian noise. The COMPAS data includes binary variables (charge degree, recidivism) and heavy-tailed count variables (priors, juvenile records), which partly violate these assumptions; LiNGAM coefficients on COMPAS are therefore linear approximations to a non-linear reality, not exact structural parameters. LiNGAM achieves high detection across the sample sizes and magnitudes tested in Section V-D, but real-world bias magnitudes are unknown and may be small. All causal conclusions rely on untestable assumptions (faithfulness, acyclicity); our results should be treated as causal hypotheses that require domain validation, legal analysis, and expert review before guiding any policy, especially in high-stakes contexts.

VIII. Conclusion

Causal discovery does not supplant correlation-based bias assessments; it complements them. Where correlational metrics reveal that a disparity exists, causal methods localize where it originates and identify which interventions could plausibly address it. Across both a controlled synthetic loan-approval dataset and the real-world COMPAS recidivism data, the LiNGAM variants—and DirectLiNGAM in particular—recovered the discriminatory pathway that constraint-based, score-based, and permutation-based methods systematically failed to detect. Algorithm choice is, therefore, not methodologically neutral: it determines which forms of discriminatory bias remain visible. We additionally identified and characterized a structurally recurring failure mode in the audit setting for the GRaSP model, in which high-R^2 outcome variables are misclassified as exogenous: a finding with practical implications for any causal-discovery workflow applied to algorithmic scores constructed from their own input features.

Future work includes extending this five-step pipeline (Fig. 6) to other high-stakes domains such as lending, hiring, and healthcare; incorporating kernel-based functional causal models (e.g., RESIT, HSIC-LiNGAM) to relax the linearity assumption; and developing regulatory-aligned reporting templates that translate causal graphs into concrete interventions for legal, policy, and engineering audiences. Distinguishing whether bias exists from understanding how it propagates is essential to designing interventions that address root causes rather than mask symptoms. Replication code for this paper is available at https://anonymous.4open.science/r/causal_bias_detection-2151 (anonymized for review).

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